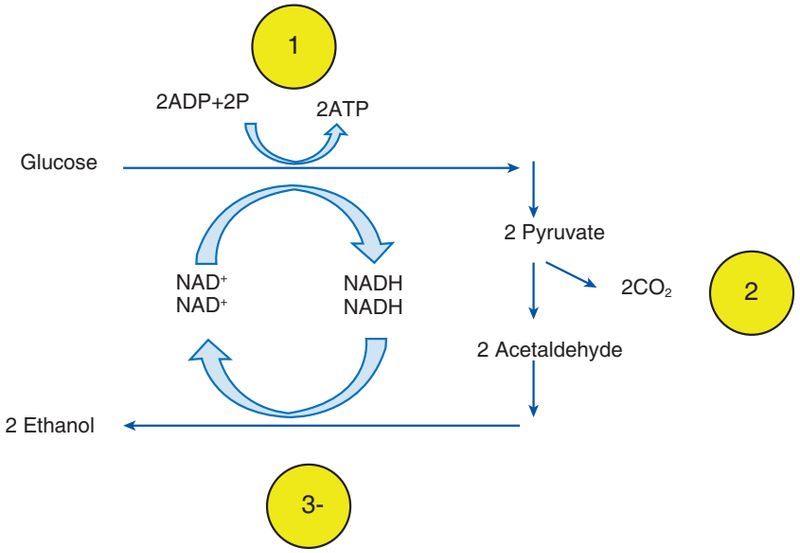


Figure 4.9: Biochemical exchange of NADH and NAD^+

(Figure 4.9). Organic electron acceptors such as pyruvate or acetaldehyde react with NADH to form NAD^+, producing CO_2 and organic solvent like ethanol. Fermentation can be classified as Lactic acid fermentation and alcohol fermentation.



Aquifex (water maker) of Aquificae is a diverse collection of bacteria that live in harsh environmental settings. These can produce water by oxidizing hydrogen.

4.9.1 Lactic acid fermentation

During Glycolysis, in the first step of lactic acid fermentation, a molecule of glucose is oxidized to 2 molecules of pyruvic acid and it generates the energy. In the next step pyruvic acid is reduced by NADH to form lactic acid. *Lactobacillus* and *Streptococcus* are some of the lactic acid producing genera (Figure 4.10).

Milk is converted into fermented products such as curd, yogurt and cheese. The fermentation of lactose in milk by these bacteria produces lactic acid which acts on milk protein to give yogurt its texture and characteristic tart flavour. Here lactase enzyme is produce by the bacteria which convert the lactose into lactic acid.



Builds up of Lactic acid in muscle cells causes muscle cramp.

Homolactic acid fermentation

In this type of fermentation, organism produces lactic acid alone. So it is referred to as homolactic fermentation.

Glucose + 2ADP + 2P ⟶ Lactic acid + 2 ATP

Heterolactic acid fermentation

In this type of fermentation, organism produces Lactic acid as well as other acids or alcohol. So it is known as hetero

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